

GeoDB-R

An R wrapper for the GEO-DB public API obtaining worldwide data on cities, counties, islands, regions and countries.

<http://geodb-cities-api.wirefreethought.com/>

<https://wirefreethought.github.io/geodb-rest-api-docs/>

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Several API endpoints to gather information:

- **Places (islands, districts, cities)** – Get all data including respective distances, populations, locations and more.
- **Regions** - More detailed data on Regions
- **Cities** – More detailed data on Cities
- **Countries** – Similar data but with more country specific details such as currency.

Most of the API's are highly user-unfriendly and difficult to use:

- Example of an endpoint used to obtain data on all inhabited Islands within a 1200 KM radius of UBC Vancouver Campus that are not within Canada or the United States:

`https://wft-geo-db.p.rapidapi.com/v1/geo/places/Q391028/nearbyPlaces?radius=1200&excludedCountryIds=Q16,Q30&minPopulation=1&distanceUnit=KM&type=ISLAND`

Some Examples:

- Regions, Places and Cities – only accepted as “Wiki data Id” values
- Locations – only accepted as a comma delimited string in ISO-6709 longitude and latitude form
- Countries – only accepted as a comma delimited list of Wiki data Id’s or codes
- Time Zones – only accepted as Wiki Data Id’s
- Types of places included in API’s – only accepted as comma delimited strings like “AMD2” for districts
- Returns confusing metadata and hard to interpret at first glance

Our Solution

- Wrap the API's into properly named functions for specific API calls providing easy to use parameters
- Locations – now accept any address, name of a place or longitude and latitude
- Country, City, Region, Place and Time Zone names – now accepted as an array or vector of actual appropriate names, or Id's
- Create separate functions or Boolean parameters for specific types of Regions or Places
- Return Data in a clean data frame
- Return chart maps for places and population levels

Examples:

- The wrapped function for the API endpoint example earlier would be:

**GEO.FindIslands.NearPlace(place = “UBC Vancouver”,
excludeCountries = c(“Canada”, “United States”), radius = 1200,
inhabited = TRUE)**

- GEO.FindCities(Regions = vector of regions names, nearLocation = name of a place, Countries = vector of countries.....)
- GEO.Distance.Between(from = name of place , to = name of place)
- GEO.FindRegions(.....)

AND MORE!

EXTRAS

- Functions to extract the population density of any named place, or any geographical area (not provided by GEO db api)
- Charts and maps for visualizing populations, densities and more

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CHARTS

